

SAC Project Report

A Study of Social Media Engagement and Psychological Outcomes

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Track: Analytics

Platform: SAP Analytics Cloud

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1. Project Introduction

1.1 Background of the Study

Social media has become a dominant force in shaping communication, lifestyle, and mental well-being across all age groups. Platforms such as Instagram, Facebook, and Twitter are widely used by **school students, college students, salaried workers, and retired individuals**, each engaging with social media for different purposes. While these platforms foster connectivity and information sharing, excessive use has been linked to **distraction, anxiety, lack of concentration, and social comparison**, raising concerns about their long-term psychological impact.

The **average time spent on social media per day** has emerged as a critical factor in understanding these outcomes. Students often face academic distraction, professionals report reduced productivity, and retired individuals may experience both companionship and feelings of inadequacy. These effects vary significantly across **age, gender, and occupation status**, making it essential to analyze demographic differences in social media usage and mental health outcomes.

To address this, the present study leverages **SAP Analytics Cloud (SAC)**, a powerful business intelligence and analytics platform, to perform data analysis on the collected dataset. SAC enables interactive visualization, linked analysis, and AI-driven insights, making it suitable for exploring complex relationships between **social media engagement patterns and mental health indicators**. By applying SAC's analytical capabilities, this study aims to uncover correlations, highlight demographic trends, and provide evidence-based insights into the broader implications of social media consumption.

1.2 Problem Statement

The widespread adoption of social media across all age groups and occupational categories has created both opportunities and challenges for mental well-being. While platforms enable communication, entertainment, and social connection, excessive engagement has been increasingly associated with negative psychological outcomes such as **distraction, anxiety, lack of concentration, and social comparison**.

Despite growing awareness, there remains limited empirical evidence on how these effects vary across **different demographic segments—school students, college students, salaried workers, and retired individuals—as well as across gender and age groups**. Furthermore, the role of **average daily time spent on social media** as a predictor of mental health outcomes is not fully understood.

This study addresses the problem by analyzing a dataset that captures demographic variables and mental health indicators, using **SAP Analytics Cloud (SAC)** for advanced visualization and correlation analysis. The central issue is to determine whether patterns of social media usage significantly contribute to mental health challenges, and to identify how these relationships differ across population segments

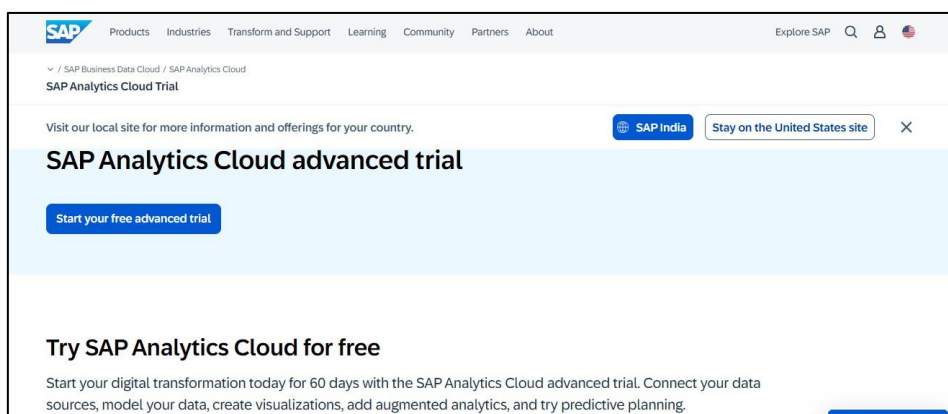
1.3 Objectives of the Project

- ❖ **To examine demographic differences** (age, gender, occupation status) in social media usage patterns.
- ❖ **To analyze the relationship** between average daily time spent on social media and mental health indicators such as distraction, anxiety, lack of concentration, and social comparison.
- ❖ **To identify high-risk groups** (e.g., students, salaried workers, retired individuals) most affected by excessive social media engagement.
- ❖ **To explore correlations** between social media usage intensity and psychological outcomes using SAC's visualization and analytical tools.

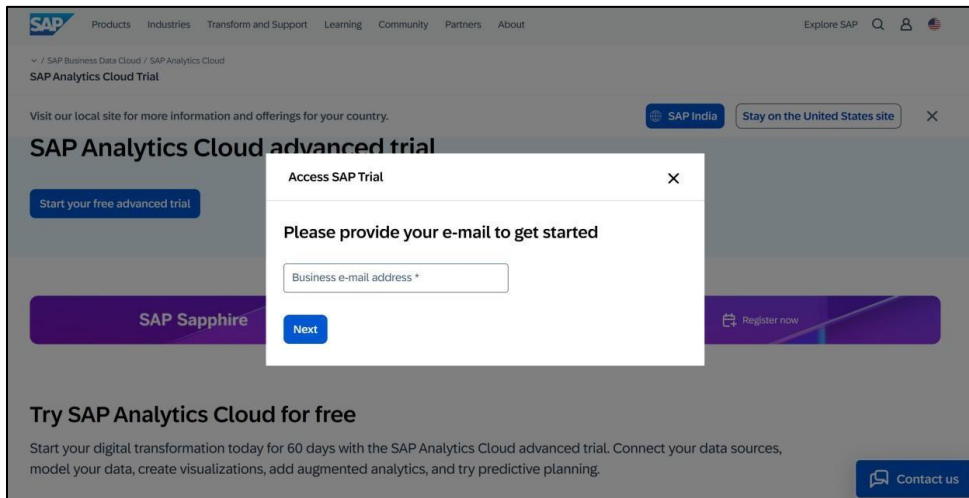
1.4 Project Setup

SAP Analytics Cloud was accessed via the SAP Analytics Cloud Advanced Trial:

- Visited <https://www.sap.com/products/data-cloud/cloud-analytics/trial.html>



- Clicked “Start Your Free Advanced Trial” and registered using an institutional email address



- Received tenant access information via email, including the Tenant URL, Username, and Password setup instructions

Tenant Access Information for SAP Analytics Cloud

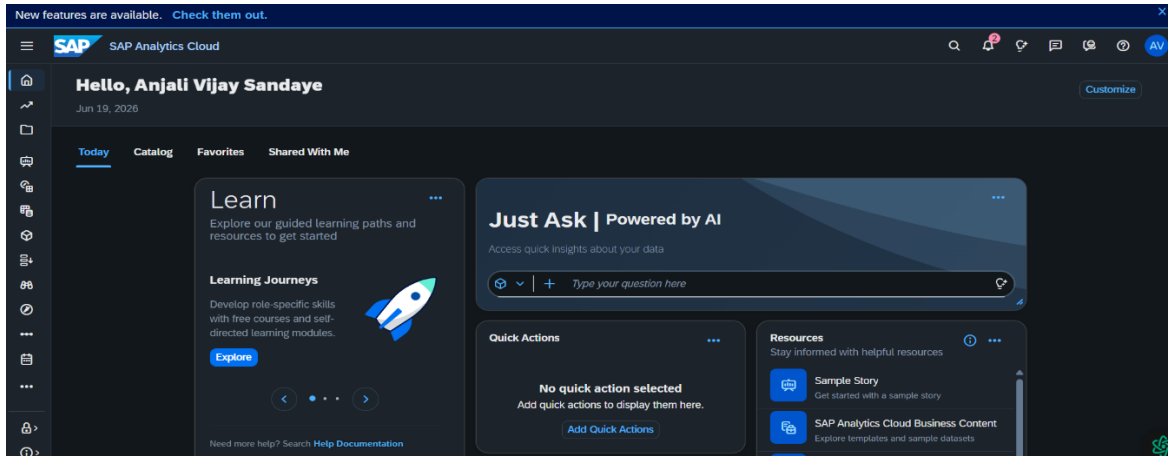
Thank you for choosing SAP Analytics Cloud. Please find below your account details to access the application.

Tenant URL <https://sactrial-sacap10-dd0mg9z2tap01u1vuvjg7wsu.ap10.hcs.cloud.sap>

Username: bansishah009@gmail.com

Password If you cannot remember your password, please click the "Forgot Password?" link on the login page. You will receive a separate email with instructions on how to set a new password. After setting your password, you can log on as described above.

- Logged into the SAC tenant and navigated to the Modeler to begin dataset import and modelling



2. Dataset Description

2.1 Data Source and Methodology

The dataset used for this study was collected from **Kaggle**, a widely recognized open-source platform for data science and machine learning projects.

Social Media v/s Mental Health dataset was selected because it provides diverse attributes that allow for a comprehensive analysis of the relationship between social media usage and mental health outcomes across different population segments.

New features are available. Check them out.											
SAP Analytics Cloud											
Hello, Anjali Vijay Sandaye											
Jun 19, 2026											
Today Catalog Favorites Shared With Me											
Learn											
Explore our guided learning paths and resources to get started											
Learning Journeys											
Develop role-specific skills with free courses and self-directed learning modules.											
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Add Quick Actions											
Resources											
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Sample Story											
Get started with a sample story											
SAP Analytics Cloud Business Content											
Explore templates and sample datasets											

2.2 Dataset Features and Attributes

The final dataset contains 470 rows and 17 columns, structured into Dimensions, and Measures.

Measures

Measure	Description
Purposeless Use of Social Media	Frequency of using social media without a clear purpose or goal.
Distraction likelihood	Level of disturbance in focus or attention caused by social media.
Difficulty in Concentration	Difficulty in maintaining sustained attention during tasks.
Restlessness	Feeling of unease or inability to stay calm, linked to social media use.
Frequency of Anxious Thoughts	How often the respondent experiences anxiety-related thoughts.
Self-Doubt	Degree of uncertainty or lack of confidence in oneself.
Depression Ratio	Extent to which respondents report depressive feelings or symptoms.
Fluctuation in Daily Activities	Variation or inconsistency in routine activities due to social media influence.
Age	Numeric value representing the respondent's age (not on 1–5 scale).
Average Time Spent	Daily average time spent on social media platforms (numeric measure).
Social Comparison Orientation	Tendency to compare oneself with others based on social media content
Issues Regarding Sleep	Problems such as difficulty falling asleep, poor sleep quality, or insomnia related to social media use.

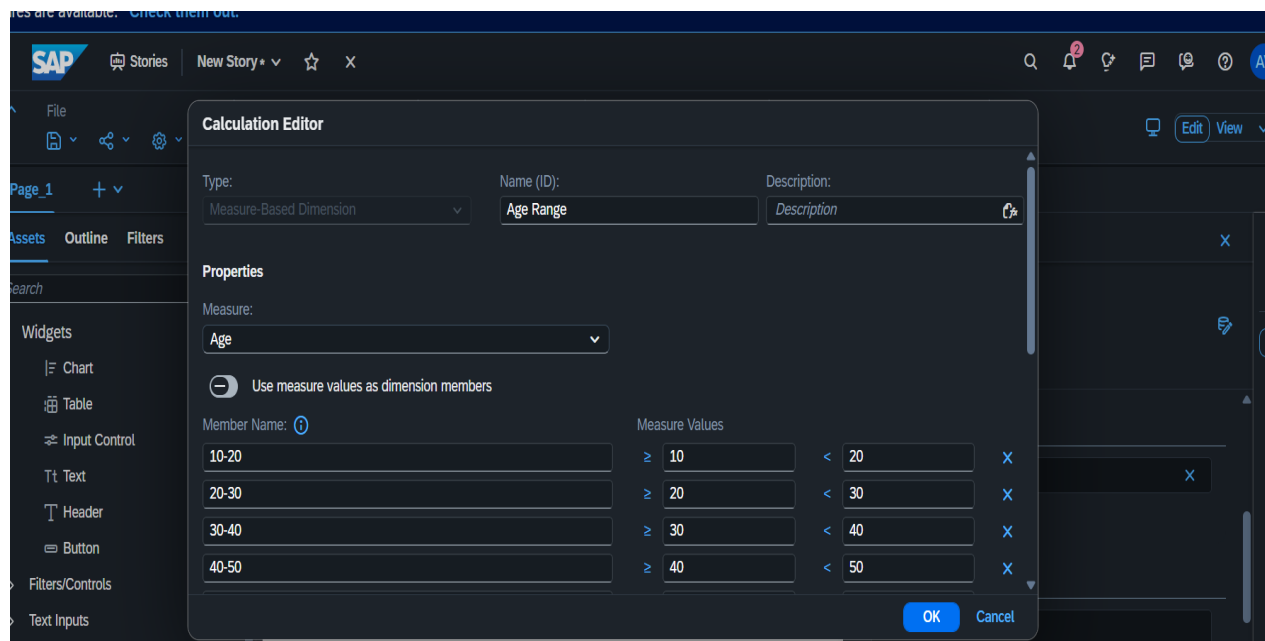
Dimensions

Dimension	Description
Relationship Status	Indicates whether the respondent is single, married, in a relationship, etc.
Gender	Identifies the respondent's gender (male, female, other).
Occupation Status	Categorizes respondents as school student, college student, salaried worker, or retired individual.
Social Media Use	Specifies whether the respondent uses social media (Yes/No).

2.3 Data Preparation and Validation

To ensure the dataset was consistent and ready for analysis in **SAP Analytics Cloud (SAC)** several preprocessing steps were applied:

- **Conversion of Age Values:** Transformed age values into whole numbers for consistency.
- **Conversion of Average Time Spent:** Converted average daily time spent on social media into whole decimal values for uniformity.
- **Deletion of Unwanted Columns:** Removed irrelevant or redundant attributes that did not contribute to the analysis.
- **Creation of Calculated Dimension – Age Range:** Designed a measure-based dimension in SAC to categorize respondents into age ranges (e.g., 10–20, 21–30, 31–40, etc.), enabling comparative analysis across demographics groups.



3. SAP Analytics Cloud (SAC) Implementation

3.1 Data Import in SAC

- The downloaded CSV dataset into SAC using Modeler -> Create Model -> Import from File (flat file / acquired data model)
- Verified the dataset loaded successfully with 470 rows and 17 columns visible in the Data Foundation preview

3.2 Data Modelling and Preparation

Key dimensions and measures established in the model:

- **Dimensions**

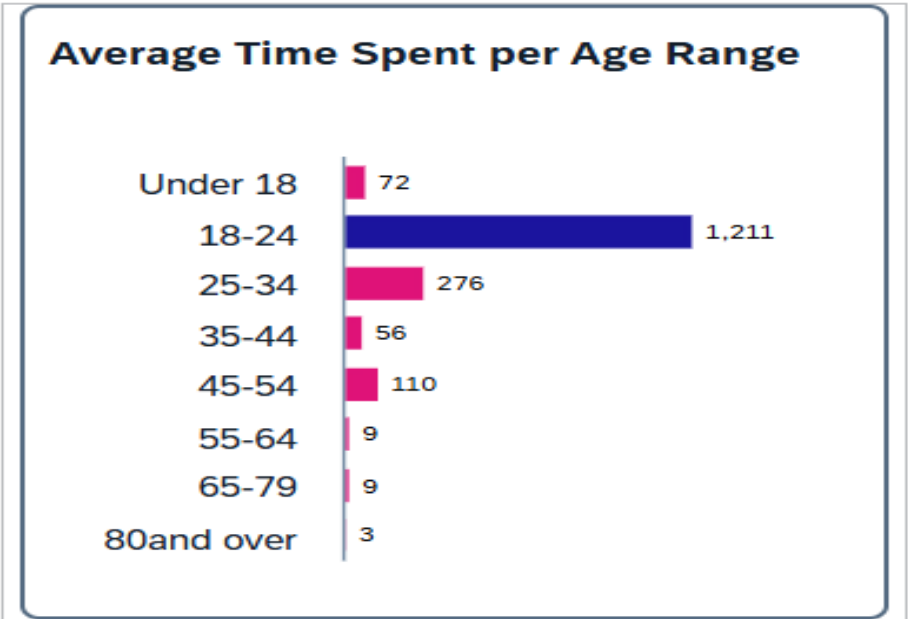
- Age Range (Under 18, 18–24, 25–34, 35–44, 45–54, 55–64, 65–79, 80+)
- Gender (Female, Male, Other)
- Occupation Status (College, School, Salaried, Retired)
- Relationship Status (Single, Married, Other)
- Social Media Use (Yes, No)

- **Measure**: Average Time Spent on Social Media (per day) , Purposeless Use of Social Media , Likelihood of Distraction, Difficulty in Concentration, Restlessness , Frequency of Anxious Thoughts , Self-Doubt , Depression Ratio , Fluctuation in Daily Activities , Social Comparison Orientation , Issues Regarding Sleep , Validation Seeking from Social Media , Age (numeric, later converted into Age Range).

3.3 Calculated Dimension

A calculated dimension was created to group respondents' ages into defined ranges. Since age was originally a numeric measure, it was transformed into a dimension using formulas that bucket values into categories

The calculated measure is used in chart visualization



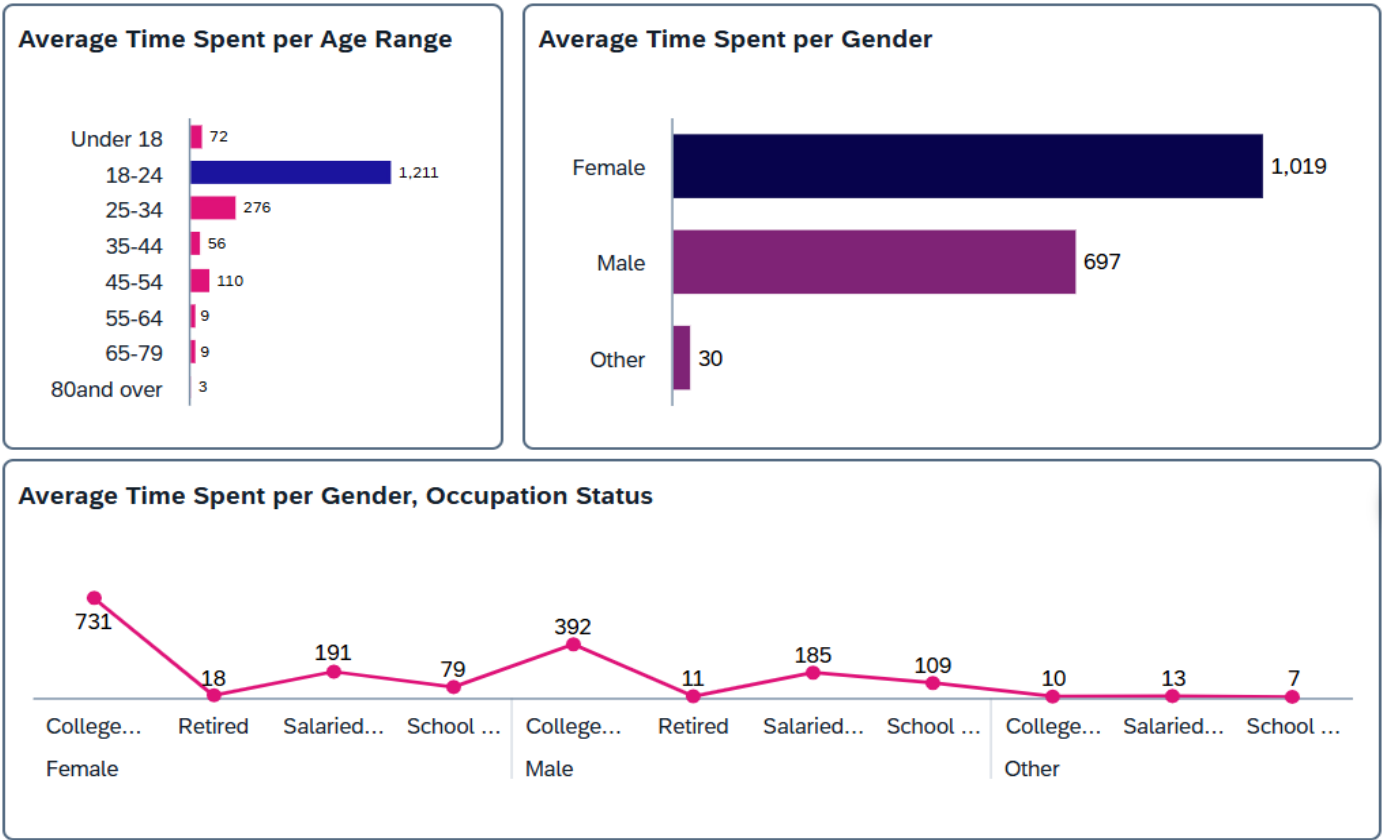
4. Dashboard Design and Development

The dashboard was designed as a Two-page SAC Story titled “A Study Social Media Engagement and Phycological Outcomes”, structured into four logical rows so that trade performance and risk exposure are visible together, addressing the core problem statement.

4.1 Demographic Analysis of Social Media Usage

The dashboard highlights how **average time spent on social media varies across demographic dimensions** such as age, gender, and occupation status.

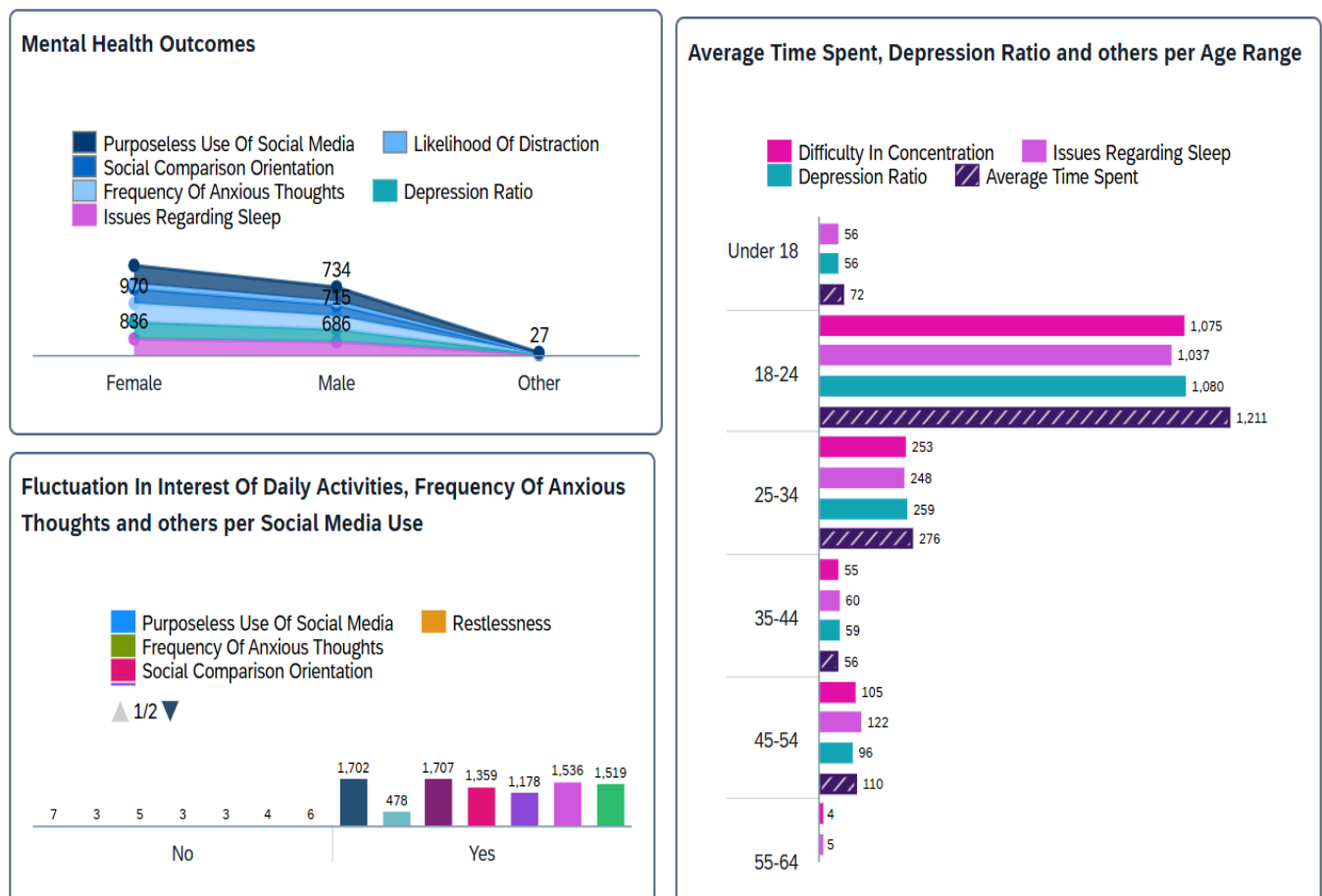
- **Age Range:** The calculated dimension *Age Range* shows that younger groups (18–24) spend significantly more time on social media compared to older groups.
- **Gender and Occupation:** Female college students recorded the highest average usage, followed by male college students, while retired and salaried groups showed comparatively lower values. This visualization makes it easy to identify **which demographic segments are most engaged with social media**.



4.2 Finding the Relation Between Average Time Spent and Mental Health

The second part of the dashboard explores the **correlation between time spent on social media and mental health indicators**.

- Respondents with higher average usage reported **greater distraction, restlessness, and issues regarding sleep**.
- Measures such as *frequency of anxious thoughts, social comparison orientation, and depression ratio* showed stronger values among heavy users compared to those with limited or no social media use.
- The grouped bar charts and KPI indicators clearly demonstrate that **increased social media consumption is linked to negative psychological outcomes**, especially among younger demographics.



5. Visualization and Features

5.1 Charts and Graphs Used

Chart Type	Usage in Dashboard
Bar Chart	Used to show Average Time Spent per Age Range and Fluctuation in Daily Activities, Anxious Thoughts, Sleep Issues, etc. per Social Media Use . Helps compare categorical values across groups.
Line Chart	Used for Average Time Spent per Gender and Occupation Status , showing trends and differences across categories.
Grouped Bar Chart	Used to compare Yes vs No social media use across multiple mental health indicators (e.g., restlessness, validation seeking, sleep issues). Highlights contrast between groups.
KPI/Indicator Card	Displays single numeric values like Likelihood of Distraction (1,597) , Frequency of Anxious Thoughts (1,713) , and Social Comparison Orientation (1,362) . Provides quick, at-a-glance metrics.
Area Chart	Used in Mental Health Outcomes by Gender , showing overlapping measures (distraction, depression ratio, sleep issues, etc.) across female, male, and other categories. Useful for multi-measure comparison.
Bubble Chart	Used for Validation Seeking from Social Media per Age Range and Gender , showing clusters and intensity of validation behavior. Highlights differences in magnitude visually.
Pie Chart (optional)	Used in some dashboards for distribution of social media use (Yes/No) , though in your case grouped bars were more prominent.

5.2 Smart Predict and Forecasting

In Sap Analytics Cloud, Smart Predict and Forecasting feature was applied to extend the insights from the dashboard beyond descriptive analytics. This allowed the project to move into predictive modeling and trend forecasting for social media usage and its impact on mental health outcomes.

- **Purpose:**

The goal was to identify **future patterns** in social media engagement and anticipate **potential psychological outcomes** such as distraction, anxious thoughts, sleep issues, and depression ratio.

- **Implementation:**

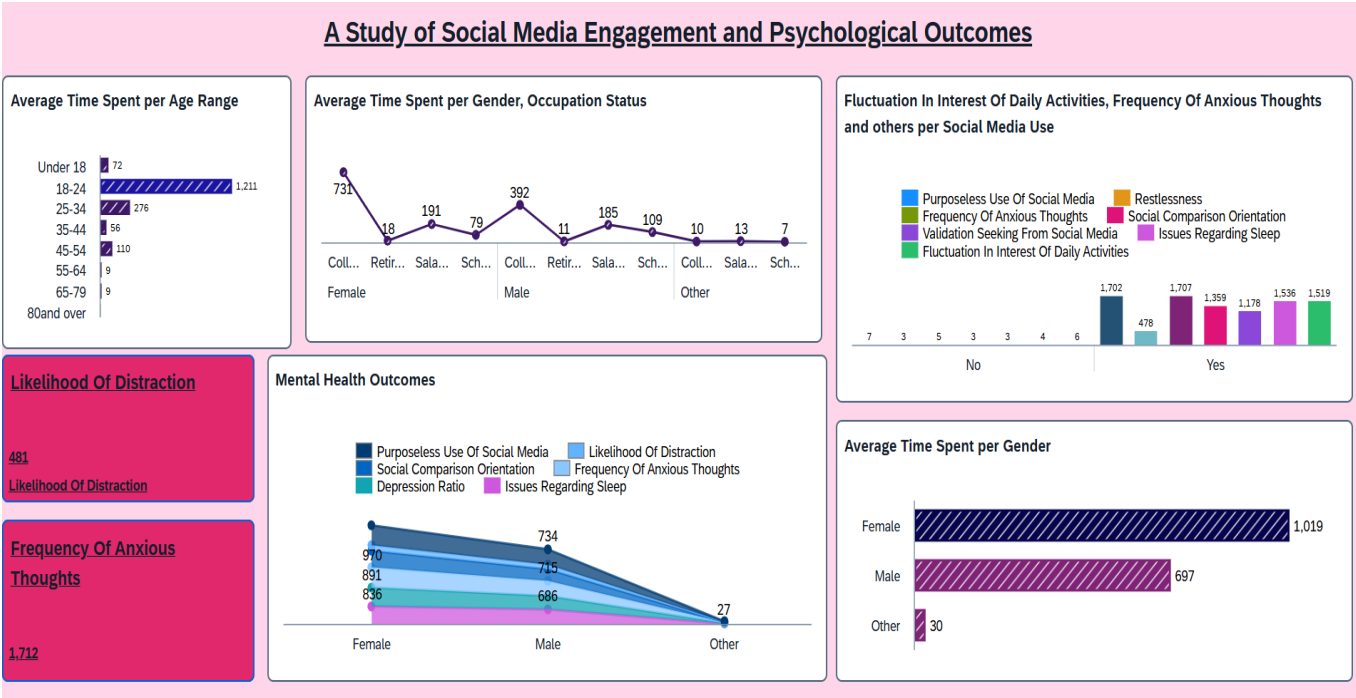
- **Predictive Models** were built using SAC’s Smart Predict to estimate how **average time spent on social media** influences mental health indicators.
- **Forecasting** was applied to project future values of measures like *frequency of anxious thoughts* and *issues regarding sleep* across different age ranges and demographics.
- **Calculated Dimensions** (e.g., Age Range) were integrated into the predictive analysis to highlight differences across demographic groups.

- **Key Insights:**

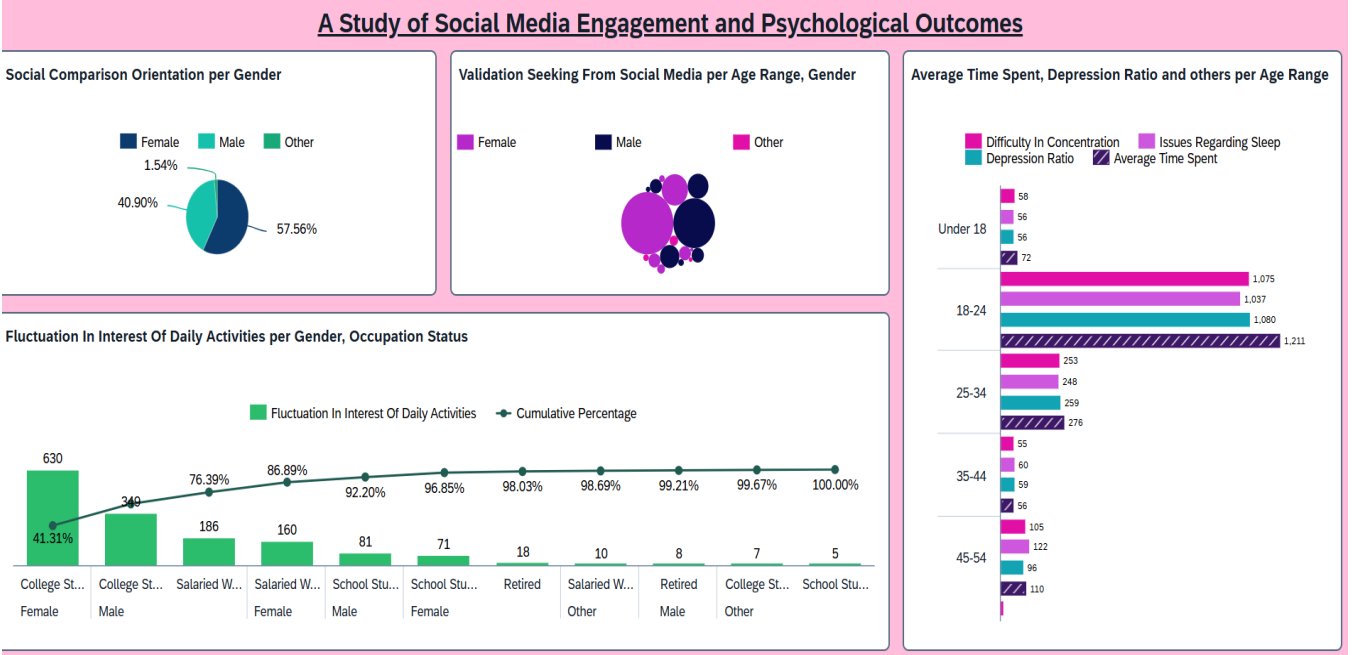
- Younger age groups (18–24) are forecasted to continue showing **higher average time spent**, which correlates with increased distraction and sleep issues.
- Occupation categories such as **college students** are predicted to remain the most vulnerable to negative psychological outcomes.
- Forecasting trends suggest that **mental health indicators rise proportionally with social media engagement**, especially in categories like *validation seeking* and *social comparison orientation*.

Feature	Usage in Dashboard
Smart Predict	Built predictive models linking average time spent on social media with mental health measures (e.g., distraction, anxious thoughts, sleep issues).
Forecasting	Projected future trends of social media usage and psychological outcomes across age ranges and demographics.
Integration with Dimensions	Used calculated dimension <i>Age Range</i> and demographic categories (gender, occupation) to refine predictions.

6. Results and Insights



Demographic Usage Patterns: Younger age groups (18–24) recorded the **highest average time spent on social media**, followed by 25–34. Female college students showed the most engagement compared to other gender–occupation categories.



- **Mental Health Correlations:** Heavy social media users reported **greater distraction, restlessness, anxious thoughts, and sleep issues** compared to non-users. The *depression ratio* and *social comparison orientation* were also significantly higher among frequent users.
- **Gender Differences:** Female respondents consistently showed **higher values across mental health indicators** (distraction, anxious thoughts, sleep issues) compared to male and other categories.
- **Validation and Comparison Behaviour:** Validation seeking and social comparison orientation were more pronounced among **younger demographics and female users**, reinforcing the link between social media engagement and psychological outcomes.

7. Conclusion

The analysis demonstrates that **social media engagement has a significant impact on mental health outcomes**, particularly among younger demographics and college students. Heavy users consistently reported higher levels of distraction, anxious thoughts, restlessness, sleep issues, and depression ratio compared to non-users. Gender differences were evident, with female respondents showing stronger correlations across most psychological indicators.

By leveraging **SAP Analytics Cloud (SAC)**, the project not only visualized these relationships but also applied **Smart Predict and Forecasting** to anticipate future trends. The findings highlight that excessive social media use is strongly linked to negative psychological outcomes, underscoring the need for awareness, balanced usage, and targeted interventions.

8. Future Enhancements

While the current dashboard provides strong insights into social media usage and mental health outcomes, several enhancements can further improve its analytical depth and usability:

- **Advanced Predictive Modelling:** Incorporate machine learning models within SAC to refine forecasts and identify hidden patterns beyond linear trends.
- **Real-Time Data Integration:** Enable live data feeds to track social media usage and psychological indicators dynamically, improving timeliness of insights.
- **Expanded Demographic Dimensions:** Add variables such as education level, income group, or geographic region to broaden the scope of analysis.
- **Interactive Storytelling Features:** Enhance dashboards with drill-down narratives, scenario simulations, and “what-if” analysis for decision-makers.
- **Benchmarking and Comparison:** Allow comparisons with external datasets (e.g., national surveys) to validate findings and provide context.

9. References

- SAP Analytics Cloud (SAC) Documentation – *Smart Predict and Forecasting* SAP Help Portal (help.sap.com in Bing)
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